

M003 Week 8 — English Worksheet

Topic: Final — My Own Museum · **Course:** 3D Coordinates · **Time:** about 60 minutes

This is your big final project. No new commands — every idea from this course working together in one build. Your museum is a **room** visitors walk into, with **flat pictures** on the walls (2D art, drawn with x and y) and **solid sculptures** on pedestals (3D art, placed with x, y, z). Plan it like a real architect: coordinates first, then build.

1 · Predict 🧠

Read each plan. Before you imagine it happening, write what you think you will see.

```
# 2D art: a picture on the back wall – z stays 0, blocks climb with x and y
set x to 0
repeat 5 times:
    place red block at (x, 1, 0)
    add 1 to x
```

Every block stays at $z = 0$; only x changes. Is this flat like a painting, or solid like a statue? From which direction do you look to see the whole picture?


```
# 3D art: a sculpture on a pedestal – all three numbers used
fill stone from (4, 0, 4) to (5, 1, 5)
place gold block at (4, 2, 4)
place gold block at (5, 2, 5)
```

This uses x , y , and z . Is it flat or solid? Can a visitor walk all the way around it? Where do the gold blocks sit — on top of the pedestal or beside it?


```
gallery floor: fill stone from (0, 0, 0) to (10, 0, 10)
pedestal A:    fill stone from (2, 0, 2) to (3, 1, 3)
pedestal B:    fill stone from (7, 0, 7) to (8, 1, 8)
```

Read the (x, z) footprints. Are the two pedestals far apart? Looking down from above, sketch in words where each one sits on the floor.

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The gallery should be a **hollow room** with a doorway. Right now it is a solid block of stone with no inside.

```
fill stone from (0, 0, 0) to (10, 4, 10)
fill air from (5, 2, 0) to (6, 3, 0)
```

Hint: before you can cut a doorway, the inside has to be hollowed out with air.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — These two sculptures should stand on **two different pedestals**. Right now both pedestals share the same (x, z) footprint, so they pile up in one spot.

```
fill stone from (2, 0, 2) to (3, 1, 3)
place gold block at (2, 2, 2)
fill stone from (2, 0, 2) to (3, 1, 3)
place blue block at (2, 2, 2)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This wall picture should be **flat 2D art** that stands up the back wall — z stays 0 while the picture climbs with x and y. Right now the second row steps **forward** onto the floor instead of **up** the wall.

```
set x to 0
repeat 4 times:
  place red block at (x, 1, 0)
  add 1 to x
set x to 0
repeat 4 times:
  place red block at (x, 1, 1)
  add 1 to x
```

Hint: to climb the wall, the row above should raise **y**, not raise z.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Blueprint Your Museum 📖

Plan **before** you build. This is the biggest build of the course, so a clear blueprint matters. Looking down from above, write the coordinates of every piece.

Gallery room: from (... , ... , ...) to (... , ... , ...) · doorway at (... , ... , ...)

Wall picture 1 — shows ... · stands on the wall where ... stays 0

Wall picture 2 — shows ... · stands on the wall where ... stays 0

Pedestal 1: from (... , ... , ...) to (... , ... , ...) · sculpture on top at (... , ... , ...)

Pedestal 2: from (... , ... , ...) to (... , ... , ...) · sculpture on top at (... , ... , ...)

4 · Build Your Museum 📷

Now switch to your homework world and build your museum from your blueprint. Check your build against this list:

- ☐ A **gallery room** with a floor, walls, and a **doorway** to walk through.
- ☐ At least **two different flat pictures** on the walls — **2D art**, where only x and y change.
- ☐ At least **two sculptures**, each on its **own pedestal** — **3D art**, using x, y, and z.
- ☐ The two pedestals **do not overlap**, and each sculpture **rests on top** of its pedestal.
- ☐ The museum matches the coordinates on your blueprint.

Send a photo or video of your finished museum, then explain what you built. Use these sentence starters — write 6 to 8 sentences total.

First, I planned ...

My gallery room is ... and its doorway is at ...

My 2D pictures are flat because ...

My 3D sculptures are solid because ...

I kept my pedestals from overlapping by ...

Each sculpture rests on its pedestal because its height starts at ...

One tricky moment was when ...

If I had more time, I would ...

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5 · Record Your Gallery Tour 📹

Take a video on your phone (or a parent's phone) and give a full **gallery tour** of your museum. Talk like a museum guide showing visitors around. Try to use these words: **museum, 2D, 3D, pedestal, coordinate.**

1. Walk in through the doorway and welcome your visitors.
2. Stop at a flat picture and say why it is **2D** — which numbers change and which stays the same.
3. Walk around a sculpture and say why it is **3D** — and which pedestal it stands on.
4. Point out how you planned the whole room with coordinates so nothing overlaps.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.